

# Research on Fault Location Algorithm for Line Protection

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**Abstract**—The algorithms of fault location by impedance, dual-end fault location and three-end fault location are introduced, and their application effects are analyzed and compared. Fault location by impedance, which uses voltage and current at one end of the line to calculate the impedance, may result in significant errors because the information of fault points can not be obtained and the influence of grounding resistance can not be overcome. The dual-end fault location, which is implemented using longitudinal current differential protection, greatly improves the accuracy of fault location because it reduces the influence of unfavorable factors such as transition resistance compared with the fault location by impedance. However neither fault location by impedance nor dual -end fault location can realize fault location of T-connected lines. In this paper, a new method, three-end fault location, for identifying the fault point occurring at the crossover point is proposed, which is easy and accurate to calculate the voltage threshold of the crossover point, and then identify the fault branch and complete the accurate fault location. The simulation test shows that this method is not affected by transition resistance and can be applied to various operation modes. Moreover, it can get accurate fault location on weak power side.

**Keywords**—line protection, fault location, transition resistance, fault location by impedance, dual-end fault location, three-end fault location

## I. INTRODUCTION

With the rapid development of national economy, the demand for electricity shows a continuous growth trend. National investment in power grids has also increased, which lead to the consequent increase in transmission line mileage. The transmission line bears the heavy responsibility of load transmission in the power grid and is also the place with the highest failure rate in the power system. Fast and accurate fault location is of great significance to power system. Especially in the case of permanent fault of line, it is necessary to find fault points quickly and accurately, then eliminate fault in time and restore power supply.

Fast and accurate fault location is the key factor to determine whether power supply can be restored quickly. Traveling wave ranging analyzes and calculates transient traveling wave signals generated by faults<sup>[1-2]</sup>. The traveling wave propagation of faults is susceptible to bus wiring mode and earth resistance, and needs equipment such as Time Synchronization System and signal source, which is costly. on the other side, it can not solve the problem of wave head reflection at the connection point when both elevated lines and cable lines are used.

Traveling wave fault location is mainly used in 500 kV lines. At present, line protection fault location is still widely used. If the accuracy of protection fault location can be improved, the investment of power grid will be greatly reduced.

At present, there are three ways for line protection fault location as following.

1) Fault location by impedance. For now there is a great amount of non-differential line protection below 110 kV voltage level, so the protection device can only collect fault electrical quantity on one side and then measure the distance by impedance. However, the ranging error is always large due to the influence of dual power supply and transition resistance<sup>[3-4]</sup>.

2) Dual-end fault location. Differential protection is the main protection for new-built stations in power grid, and the electrical quantity information can be obtained on both sides, so The algorithm of dual-end fault location has been widely used. Fault analysis of electrical quantities on both sides of the line can reduce the impact of transition resistance, thus more accurate, simple and fast fault location can be achieved<sup>[5-7]</sup>.

3) Three-end fault location. When the line is T-connected line, due to the shunt phenomenon, the dual-end fault location can not meet the requirements<sup>[8]</sup>. Three-end fault location has to be used. Three-end fault location can adapt to various operation modes and is easy to calculate. According to the three-side voltage, current and positive sequence impedance parameters of line, the T- point voltage is calculated on each side, and the fault branch is identified according to the calculated voltage amplitude of T-point. Then, the fault location at both ends is realized by using T-point and the electrical quantity at the head of the fault branch.

T-connected lines are being used widely. SGCC carried out T-connected line protection detection for the first time during 2019-2020. CSC-163T line protection device used a new method to identify fault points occurring at the crossover point. It can accurately calculate the voltage threshold at the crossover point, reliably identify fault branches and realize accurate fault location of T-connected lines. It has passed the detection successfully.

## II. FAULT LOCATION BY IMPEDANCE

The algorithm of fault location by impedance is to collect the fault voltage and fault current at the protection installation place through the protection device at the time of fault, and calculate the fault circuit impedance through

the fault voltage and fault current. The distance from the protective installation to the fault point is obtained according to the length of the line proportional to the impedance. Figure 1 is a schematic diagram of single-end fault location, assuming M-side is the ranging side.

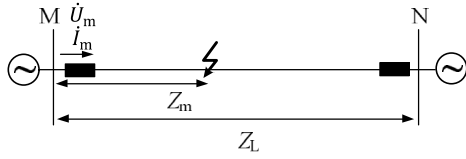


Fig. 1. Schematic diagram of single-end fault location

Without considering the transition resistance, it can be simply assumed that the measured impedance is:

$$Z_m = \frac{\dot{U}_m}{\dot{I}_m} \quad (1)$$

The distance from the protective installation place to the fault point is:

$$LEN_{mf} = \left| \frac{Z_m}{Z_L} \right| \quad (2)$$

Where:  $LEN_{mf}$  is the distance from the protection installation place to the fault point,  $Z_L$  is the positive sequence impedance per kilometer of the line, and  $Z_m$  is the positive sequence impedance from the short circuit point to the protection installation place.

However, in the actual field operation process, it is unlikely to be so idealized. In addition to the influence of PT and CT inherent error factors, there are also adverse factors such as transition resistance, incomplete symmetry of lines, boost current brought by power supply, and draw current caused by load, which lead to large measurement error. This may result in the above formula not being true. The expression of the measured impedance is as follows:

$$Z_m = Z_L L_{mf} + \frac{\dot{I}_f}{\dot{I}_m} R_f \quad (3)$$

Where:  $\dot{I}_f$  is the short-circuit current at the fault point and  $R_f$  is the positive impedance of the transition resistance at the fault point.

The  $\dot{I}_f R_f / \dot{I}_m$  in the equation is the biggest factor for the existence of errors. When the power supply is unilateral, if  $\dot{I}_f = \dot{I}_m$  The error of ranging is resistive. If the power supply is on both sides, the ranging error is not only affected by the transition resistance, but also by the ratio  $\dot{I}_f / \dot{I}_m$  of the short circuit current at the fault point and the measured current vector at the protection installation place. Since this method only needs to collect the electrical quantity on one side, and the algorithm and implementation method are relatively simple, its application is relatively wide, but the drawbacks are obvious.

### III. DUAL-END FAULT LOCATION

With the development of science and technology, the cost of optical fiber communication has dropped dramatically and the optical fiber utilization rate of line protection has been improved continuously. At present, the two major power grid companies have put forward

plans for optical fiber transformation, and it is expected that in recent years, the protection of optical fiber rate of 220 kV and above line will reach 100%. Therefore, the dual-end fault location method which collects the electrical quantity on both sides will become necessary for the line protection.

Figure 2 is a sketch of the dual-end fault location. Taking M side as an example, it is necessary to receive analog data such as the fault current and fault voltage on the N side at the time of failure through the optical fiber channel. Then accurate fault location will be obtained by calculating collected electrical quantity data on both sides.

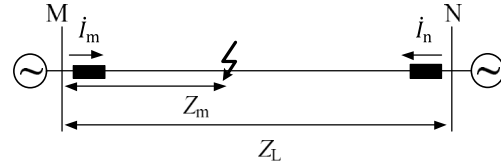


Fig. 2. Schematic diagram of the dual-end fault location

The bus voltages on both sides of M and N pass the line voltage drop from the bus to the fault point, the voltage at the fault point should be the same. Thus the following equation can be obtained:

$$\dot{U}_m - \dot{I}_m Z_m = \dot{U}_n - \dot{I}_n (Z_L - Z_m) \quad (4)$$

Where:  $Z_L$  is the positive sequence impedance of the whole line.

For Eq. (4), the following examples illustrate the values of two typical faults:

Phase A ground fault:

$$\begin{cases} \dot{U}_m = \dot{U}_{ma}, & \dot{I}_m = \dot{I}_{ma} + 3K\dot{I}_{m0} \\ \dot{U}_n = \dot{U}_{na}, & \dot{I}_n = \dot{I}_{na} + 3K\dot{I}_{n0} \end{cases} \quad (5)$$

Where  $K$  is the zero-sequence current compensation coefficient.

AB interphase fault:

$$\begin{cases} \dot{U}_m = \dot{U}_{ma} - \dot{U}_{mb}, & \dot{I}_m = \dot{I}_{ma} - \dot{I}_{mb} \\ \dot{U}_n = \dot{U}_{na} - \dot{U}_{nb}, & \dot{I}_n = \dot{I}_{na} - \dot{I}_{nb} \end{cases} \quad (6)$$

After the equation transformation, we can get:

$$Z_m = \frac{\dot{U}_m - \dot{U}_n + \dot{I}_n Z_L}{\dot{I}_m + \dot{I}_n} \quad (7)$$

The result of distance measurement is obtained from equation (2).

The calculated impedance according to Eq. (7) is the line impedance from the protection installation place to the fault point. The transition resistance is successfully eliminated, which effectively reduces the influence of the transition resistance on the ranging when the ground fault occurs.

### IV. THREE-END FAULT LOCATION

For the T-connected transmission line, as shown in Figure 3, the fault branch must be determined at first. According to the positive sequence equivalent circuit, the voltage at the crossover point (O-point) calculated by the

data of each side during normal operation is equal, but the O-point voltage calculated by the data of fault branch is different from the other two non-fault branches when the fault occurs, so the fault branch can be identified.

The operation of the system is flexible and changeable, and there are different operation modes in different periods. The operation mode of T- connected line switches between three-end operation mode and dual-end operation mode, which are explained separately as following.

#### A. Three-end Operation mode

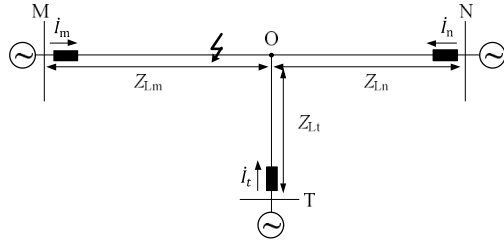


Fig. 3. Schematic diagram of T-connected line (three-end operation)

##### 1) Determination of the faulty branch

The voltage at O-point can be calculated and derived according to the voltage and current on each side:

$$\begin{cases} \dot{U}_{mo} = \dot{U}_m - \dot{I}_m Z_{Lm} \\ \dot{U}_{no} = \dot{U}_n - \dot{I}_n Z_{Ln} \\ \dot{U}_{to} = \dot{U}_t - \dot{I}_t Z_{Lt} \end{cases} \quad (8)$$

The difference of voltage on each side can be obtained by subtraction:

$$\begin{cases} \Delta \dot{U}_{mno} = |\dot{U}_{mo} - \dot{U}_{no}| \\ \Delta \dot{U}_{nto} = |\dot{U}_{no} - \dot{U}_{to}| \\ \Delta \dot{U}_{tmo} = |\dot{U}_{to} - \dot{U}_{mo}| \end{cases} \quad (9)$$

When the line is operating normally and the fault occurs at the O-point, the calculated O-point voltages on each side are equal, i.e.  $\Delta \dot{U}_{mno} = \Delta \dot{U}_{nto} = \Delta \dot{U}_{tmo}$ . However, because of the transmission error of PT and CT, the accuracy error of AD and the calculation error of software, the three differences can not be completely equal, so it is necessary to find a suitable threshold  $U_{Limit}$ . When the three voltage differences are less than  $U_{Limit}$ , it can be determined that the fault occurs at the O-point.

Through a large number of experiments, it is found that the threshold value of  $U_{Limit}$  in Eq. (10) is ideal.

$$U_{Limit} = \max \{1V, \max(U_{mo}, U_{no}, U_{to}) / 20\} \quad (10)$$

If the above three voltage differences are not all less than  $U_{Limit}$ , the fault branch can be identified by the method that the O-point voltage calculated by the data of fault branch differs from the O-point voltages calculated by the data of other two non-fault branches:

- If  $\Delta \dot{U}_{mno}$  is the smallest, the fault point is at branch TO.
- If  $\Delta \dot{U}_{nto}$  is the smallest, the fault point is at branch MO.

- If  $\Delta \dot{U}_{tmo}$  is the smallest, the fault point is at branch NO.

##### 2) Fault location

When the fault point is at O-point through calculations, the fault location on each side is the length of the respective branch.

When the fault point is not at O-point, taking the fault point on the MO branch as an example. The voltage and current at O-point are calculated from the non-faulty branches NO and TO.

$$\begin{cases} \dot{U}_o = \frac{\dot{U}_{no} + \dot{U}_{to}}{2} \\ \dot{I}_o = \dot{I}_n + \dot{I}_t \end{cases} \quad (11)$$

M-point and O-point can be thought of as two ends of a line. So the fault electrical quantity of M-point and the electrical quantity of O-point by calculated can be obtained. Then the distance  $CJM$  from the fault occurrence point to the M-side protection installation can be easily calculated by formula (7). The ranging result of N-side and T-side can also be obtained, as the N-side ranging results  $CJN = LenM - CJM + LenN$ , and the T-side ranging results  $CJT = LenM - CJM + LenT$ . If faults occur in other branches, the fault location results can be got by the same way.

#### B. Dual-end Operation mode

Figure 4, taking the T-side maintenance and switch disconnection as an example, carries out the dual-end fault location based on the current and voltage on M-side and N-side (at this time, the line length is  $LenM + LenN$ ). If  $(CJM + CJN) > (LenM + LenN)$ , the fault occurs on the maintenance branch. It is still necessary to calculate the voltage at O-point ( $\dot{U}_o = (\dot{U}_{mo} + \dot{U}_{no}) / 2$ ) and the current at O-point ( $\dot{I}_o = \dot{I}_m + \dot{I}_n$ ). Since the T-side switch is off and the current is essentially zero, the impedance distance can only be measured by the current and voltage at O-point when the fault occurs. The distance results of M side and N side can be obtained respectively by calculating the distance  $CJO$  between the fault point and O-point, The M side distance result should be  $CJM = LenM + CJO$ , and the N side distance result should be  $CJN = LenN + CJO$ .

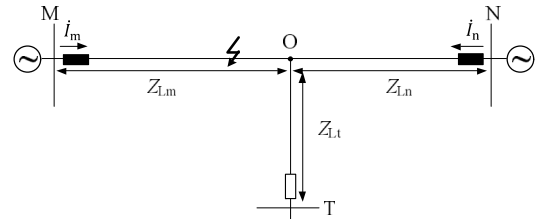


Fig. 4. Schematic diagram for T-connected line (running at dual-end)

## V. SIMULATION TEST

### A. Simulation Test of Three-End

To verify the performance of the fault location for three-end line differential protection proposed in this paper, a 110 kV simulation model is built on RTDS (real time digital simulation system) platform according to Figure 5 for a three-end T-connection system. The distance of M-side, N-side and T-side from O-point are  $L_m$  (60km),  $L_n$  (90km) and  $L_t$  (40km) respectively.  $F_1$ ,  $F_2$  and  $F_3$  are outlet faults of each branch line,  $F_4$  is T-point fault,

$F_5$ ,  $F_6$  and  $F_7$  are midpoint faults of each branch line respectively.

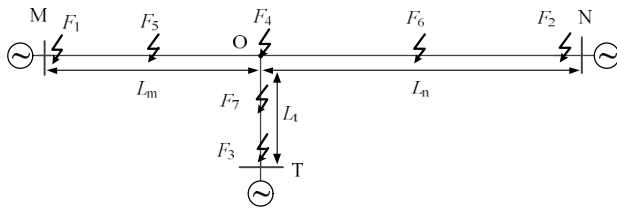


Fig. 5. Wiring Diagram of RTDS Simulation System (three-end system)

According to the requirements of GB / T 26864-2011 the dynamic test of power system protective products, a complete simulation test is carried out on the three-end system, and different operation modes with the power supply and the weak feeder on each side are built to simulate metal faults and different transition resistance faults, so as to verify the accuracy of fault location of current differential protection.

Table 1 shows the operation mode when all sides are power supplies. The faults  $F_1$ ,  $F_2$ ,  $F_3$  and  $F_4$  are simulated metal faults, and the faults  $F_5$ ,  $F_6$  and  $F_7$  are simulated faults through 100  $\Omega$  transition resistance. In Table 2, M is the power supply side, N and T are the operation mode of weak feed side, also simulating metal faults.

TABLE I. SIMULATION RESULTS OF FAULT LOCATION WITH THREE-END OF POWER SUPPLY

| Fault point | Fault type      | M-side ranging | N-side ranging | T-side ranging |
|-------------|-----------------|----------------|----------------|----------------|
| $F_1$       | AN              | 0.1 km         | 149.0 km       | 99.5 km        |
| $F_2$       | AB              | 149.0 km       | 0.2 km         | 130.5 km       |
| $F_3$       | ABN             | 99.5 km        | 128.5 km       | 0.1 km         |
| $F_4$       | ABC             | 60.0 km        | 90.0 km        | 40.0 km        |
| $F_5$       | AN-100 $\Omega$ | 29.6 km        | 120.5 km       | 70.5 km        |
| $F_6$       | BN-100 $\Omega$ | 105.5 km       | 44.7 km        | 85.5 km        |
| $F_7$       | CN-100 $\Omega$ | 80.5 km        | 108.5 km       | 19.8 km        |

TABLE II. SIMULATION RESULTS OF FAULT LOCATION OF THREE-END OF WEAK FEEDER LINE

| Fault point | Fault type | M-side ranging | N-side ranging | T-side ranging |
|-------------|------------|----------------|----------------|----------------|
| $F_1$       | BN         | 0.0 km         | 150.0 km       | 100.0 km       |
| $F_2$       | BC         | 148.0 km       | 0.8 km         | 128.0 km       |
| $F_3$       | CAN        | 99.0 km        | 129.0 km       | 0.6 km         |
| $F_4$       | CN         | 58.5 km        | 91.0 km        | 41.8 km        |
| $F_5$       | AB         | 29.0 km        | 119.5 km       | 70.0 km        |

### B. Simulation Test of Ranging at Dual-end

Figure 6 shows the simulation model of 110 kV dual-end system. The total length of MN line is 100 km, and O-point is 60 km away from side M and 40 km away from side N.  $F_1$ ,  $F_2$  and  $F_3$  are the outlet faults of side M, O-point and side N respectively. The metal faults and the faults through different transition resistors are simulated respectively by simulating the operation modes such as power supply on both sides and weak feed on one side, so as to verify the accuracy of dual-end fault location. Some results of the test are shown in Table 3.

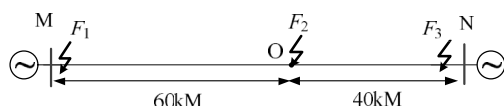


Fig. 6. RTDS simulation system wiring diagram (dual-end system)

TABLE III. FAULT LOCATION SIMULATION OF DUAL-END SYSTEM

| Operation mode             | Fault point | Fault type      | M-side ranging | N-side ranging |
|----------------------------|-------------|-----------------|----------------|----------------|
| Power supply on both sides | $F_1$       | AN              | 0.1 km         | 99.9 km        |
|                            | $F_2$       | AB              | 60.3 km        | 39.7 km        |
|                            | $F_3$       | ABN             | 99.9 km        | 0.1 km         |
| Power supply on both sides | $F_1$       | AN-100 $\Omega$ | 0.3 km         | 99.7 km        |
|                            | $F_2$       | BN-100 $\Omega$ | 56.0 km        | 44.0 km        |
|                            | $F_3$       | CN-100 $\Omega$ | 97.5 km        | 2.5 km         |
| N-side weak feed           | $F_1$       | CN              | 0.1 km         | 99.9 km        |
|                            | $F_2$       | CA              | 58.2 km        | 41.8 km        |
|                            | $F_3$       | ABC             | 98.5 km        | 1.5 km         |

### C. Analysis of Simulation Test Results

According to the ranging results in Table 1, Table 2 and Table 3, the error relative to the total length of each line is calculated, and the ranging error is not more than 5%. The test results show that under different operation modes, the fault location algorithm proposed in this paper has high accuracy and can meet the field requirements.

## VI. CONCLUSION

The fault location, as an auxiliary function of line protection, has attracted more and more attention. With the optical fiber rate of line protection getting higher and higher, at present, the mode of using dual-end fault location as the main method and fault location by impedance as the auxiliary method has been realized. Three-end fault location is a further application based on dual-end fault location. In this paper, the proposed new method for accurate calculation of crosspoint voltage threshold by the line protection device, and the method to determine fault branch by calculating the O-point voltages of three branches, which shows the O-point voltage calculated by the data of fault branch is different from that calculated by the data of the other two non-fault branches, can realize accurate fault location of T-connected lines. This method has achieved good results in field application.

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